

Statement of Purpose

Yogesh Kamboj | PhD Application | Centre for Traditional Indian Knowledge and Skills (CT)

Topics of Interest: Sanskrit Computational Linguistics; Pāṇinian Grammar in Neural NLP; Sandhi, Samāsa, and Kāraka Analysis with Aṣṭādhyāyī Sūtra Citation

Vāgarthāviva saṃprktau vāgarthapratipattaye | Jagataḥ pitarau vande pārvatīparamēśvarau ||

Kālidāsa's invocation of the inseparable union of *vāk* and *artha* names the problem I have been working on: making the hidden grammatical meaning behind Sanskrit word transformations computationally visible. Given the sandhied input *hanti naḥ rāmarūpeṇa yadi vā svayam antakaḥ*, my system Sūtradīpa produces:

hanti naḥ [8.2.66>6.1.114>6.1.87] rāma [compound] rūpeṇa yadi vā svayam antakaḥ

Each bracket cites the specific Aṣṭādhyāyī sūtra governing the sandhi at that boundary. No published system (surveyed: 45+) produces sūtra citations alongside segmentation. On the SIGHUM benchmark, Sūtradīpa achieves 94.01% perfect match on clean test sentences, matching or slightly exceeding TransLIST (93.96%) and ByT5-Sanskrit (93.83%). The system was built using the DCS-SH Alignment dataset (Krishnan, Kulkarni & Huet, 2025), 2,548 sandhi rules from the Samsādhani system (University of Hyderabad), and ByT5-Sanskrit (Nehrdich et al., EMNLP 2024). My contribution is the novel pipeline design, task formulation, and experimental methodology — not the underlying tools, which I gratefully acknowledge.

Separately, for my M.Tech dissertation at DIAT Pune, I fine-tuned ByT5-Sanskrit for Named Entity Recognition on the Mahānāma dataset (EMNLP 2025), achieving F1 = 0.860 across 15 systematic experiments — the first comprehensive NER baselines for Sanskrit. Both projects are in their final stages; I want to bring them to proper conclusion.

Justification of academic background. I come from Information Technology and Computer Science, not Sanskrit. My M.Tech coursework in Machine Learning, Optimization, and Simulation gave me the computational tools; what I lack is deep knowledge of the Pāṇinian grammatical tradition. The sūtra labels in Sūtradīpa come from a deterministic tool, not from a scholar — I

cannot personally verify whether every sūtra chain is correct. This is the most important limitation of my work, and it is why I need guidance from someone who knows the Aṣṭādhyāyī deeply.

My path to Sanskrit began with an NPTEL course on Knowledge Representation by Prof. Deepak Khemani, but what deepened it was encountering the concept of *vāk śuddhi* and its connection to *vicāra śuddhi* — that disciplining speech disciplines thought. The Aṣṭādhyāyī is the most rigorous system of *vāk śuddhi* ever created. I came to believe that between the two formal systems humanity has built to understand the world — mathematics and Sanskrit grammar — Sanskrit as a whole is more developed in holding *meaning* in mind whose formal symbol has not yet been discovered or defined. That conviction is what drives me.

Proposed research. Extending from sandhi to samāsa (2.1.x-2.2.x sūtras) and kāraka (1.4.x series) — teaching neural models to cite progressively deeper layers of Pāṇinian grammar. I know enough to see these are real problems; I do not know enough to solve them alone. These are the directions of a beginner — if properly guided, I will follow that guidance.

Why CTIKS and Prof. Malhar Kulkarni. His Computational Aṣṭādhyāyī project and dual presence in Pāṇinian scholarship and CFILT make him the ideal mentor for this intersection. I am open to whatever direction he considers most important.

I pursue this not only as research but as personal growth — I believe that as I develop through this disciplined study, I grow in my capacity to contribute to the world. The pursuit of understanding is both the method and the purpose. I am willing to face whatever difficulty that requires.